Number Logic

Year 4 · **Lesson Worksheet**

Worked Examples

Example 1: 6 and 2 go together to make 8. 9 and 4 go together to make 13. Using the same rule, what do 7 and 5 make?

Answer: 12 — the rule is: add the two numbers together (6+2=8, 9+4=13, so 7+5=12).

Example 2: Four of these five numbers share a property, but one does not belong: 10, 15, 20, 24, 30. Which number is the odd one out, and why?

Answer: 24 is the odd one out — the other four (10, 15, 20, 30) are all multiples of 5, but 24 is not.

Example 3: Ben is 6 years old. Mia is twice as old as Ben. Jack is 3 years older than Mia. How old is Jack?

Answer: 15 years old — Mia is 2×6 = 12, and Jack is 12+3 = 15.

Purple — Accessible (15 questions)

1. Four of these five numbers share a property, but one does not belong: 15, 20, 25, 30, 32. Which number is the odd one out, and what property do the other four share?
2. 3 and 5 go together to make 8. 10 and 2 go together to make 12. Using the same rule, what do 6 and 9 make?
3. Cara is 4 years old. Beth is twice as old as Cara. Amir is 3 years older than Beth. How old is Amir?
4. 2 and 9 go together to make 11. 6 and 6 go together to make 12. Using the same rule, what do 7 and 8 make? A) 13 B) 14 C) 15 D) 16 E) 17
5. Four of these five numbers share a property, but one does not belong: 12, 16, 18, 21, 24. Which number is the odd one out, and what property do the other four share?
6. Sam scored 9 points in a quiz. Priya scored 6 more points than Sam. How many points did Priya score?
7. In this number grid, each row follows the same rule. Row 1: 3, 4, 7. Row 2: 5, 6, 11. Row 3: 8, 2, ?. What is the missing number?
8. 4 and 4 go together to make 8. 7 and 6 go together to make 13. Using the same rule, what do 8 and 9 make?
9. Priya is older than Sam. Sam is older than Tom. Select ALL statements that must be true. A) Priya is older than Tom B) Tom is the youngest of the three C) Sam is older than Priya D) Priya is the oldest of the three E) Tom is older than Sam
10. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 20 B) 30 C) 40 D) 45 E) 50
11. Finn has 5 marbles. Grace has three times as many marbles as Finn. How many marbles does Grace have?
12. 11 and 2 go together to make 13. 5 and 9 go together to make 14. Using the same rule, what do 10 and 6 make?
13. Tom is 7 years old. Lily is 2 years older than Tom. Noah is 4 years younger than Lily. How old is Noah? A) 3 B) 4 C) 5 D) 6 E) 7

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14. Four of these five numbers share a property, but one does not belong: 9, 12, 15, 17, 21. Which number is the odd one out, and what property do the other four share?

15. 12 and 3 go together to make 15. 8 and 8 go together to make 16. Using the same rule, what do 9 and 10 make? A) 17 B) 18 C) 19 D) 20 E) 21

Green — Standard (15 questions)

1. Four of these five numbers share a property, but one does not belong: 16, 25, 36, 40, 49. Which number is the odd one out, and what property do the other four share?

2. 9 and 4 go together to make 5. 15 and 8 go together to make 7. Using the same rule, what do 20 and 12 make?

3. Liam has 8 stickers. Zara has 5 more stickers than Liam. Omar has twice as many stickers as Zara. How many stickers does Omar have?

4. 5 and 1 go together to make 11. 2 and 6 go together to make 10. Using the same rule, what do 4 and 5 make? A) 11 B) 12 C) 13 D) 14 E) 15

5. Four of these five numbers share a property, but one does not belong: 12, 16, 20, 22, 28. Which number is the odd one out, and what property do the other four share?

6. Priya scored 72 in a test. Dev scored 10 more than Priya. Asha scored half of Dev's score. How many marks did Asha score?

7. In this number grid, each row follows the same rule. Row 1: 3, 2, 8. Row 2: 5, 4, 14. Row 3: 6, 1, ?. What is the missing number?

8. 3 and 5 go together to make 11. 4 and 2 go together to make 10. Using the same rule, what do 6 and 3 make?

9. 11 and 5 go together to make 6. 20 and 13 go together to make 7. Using the same rule, select ALL of these pairs that would make 8. A) 19 and 11 B) 15 and 6 C) 23 and 15 D) 10 and 3 E) 30 and 22

10. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 12 B) 21 C) 24 D) 29 E) 33

11. Table A sold 18 cakes at a bake sale. Table B sold three times as many cakes as Table A. Table C sold 15 fewer cakes than Table B. How many cakes did Table C sell?

12. 20 and 14 go together to make 6. 17 and 9 go together to make 8. Using the same rule, what do 25 and 16 make?

13. Noah is 10 years old. His sister Ivy is 2 years younger than Noah. Their cousin Max is 5 years older than Noah. How old is Max? A) 13 B) 14 C) 15 D) 16 E) 17

14. Four of these five numbers share a property, but one does not belong: 11, 13, 17, 19, 21. Which number is the odd one out, and what property do the other four share?

15. 18 and 11 go together to make 7. 30 and 19 go together to make 11. Using the same rule, what do 26 and 9 make? A) 15 B) 16 C) 17 D) 18 E) 19

Red — Challenge (15 questions)

1. Four of these five numbers share a property, but one does not belong: 12, 18, 24, 30, 32. Which number is the odd one out, and what property do the other four share?

2. 5 and 3 go together to make 7. 6 and 2 go together to make 10. Using the same rule, what do 8 and 5 make?

3. Team A collected £40 for charity. Team B collected £20 more than Team A. Team C collected half as much as Team B. How much did Team A and Team C collect together?

4. 2 and 3 go together to make 7. 4 and 1 go together to make 9. Using the same rule, what do 5 and 2 make? A) 10 B) 11 C) 12 D) 13 E) 14

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5. Four of these five numbers share a property, but one does not belong: 14, 21, 28, 35, 38. Which number is the odd one out, and what property do the other four share?

6. Omar is 8 years old. His sister Priya is 2 years older than Omar. Their uncle is 4 times as old as Priya. How old is their uncle?

7. In this number grid, each row follows the same rule. Row 1: 4, 3, 11. Row 2: 5, 2, 12. Row 3: 6, 5, ?. What is the missing number?

8. 4 and 6 go together to make 20. 8 and 3 go together to make 22. Using the same rule, what do 7 and 5 make?

9. Mia is older than Noah. Noah is the same age as Omar. Omar is older than Priya. Select ALL statements that must be true. A) Mia is older than Omar B) Noah is older than Priya C) Omar is the youngest of the four D) Priya is older than Noah E) Mia is the oldest of the four

10. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 24 B) 36 C) 42 D) 44 E) 48

11. Jo has 8 sweets. Kira has twice as many sweets as Jo. Leo has 6 fewer sweets than Kira. How many sweets do Kira and Leo have together?

12. 5 and 3 go together to make 11. 6 and 4 go together to make 14. Using the same rule, what do 7 and 5 make?

13. Row A scored 6 points. Row B scored 4 times as many points as Row A. Row C scored 7 more points than Row B. How many points did Row C score? A) 24 B) 27 C) 31 D) 34 E) 37

14. Four of these five numbers share a property, but one does not belong: 9, 27, 36, 54, 58. Which number is the odd one out, and what property do the other four share?

15. 3 and 4 go together to make 14. 5 and 2 go together to make 14. Using the same rule, what do 6 and 3 make? A) 16 B) 17 C) 18 D) 19 E) 20

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